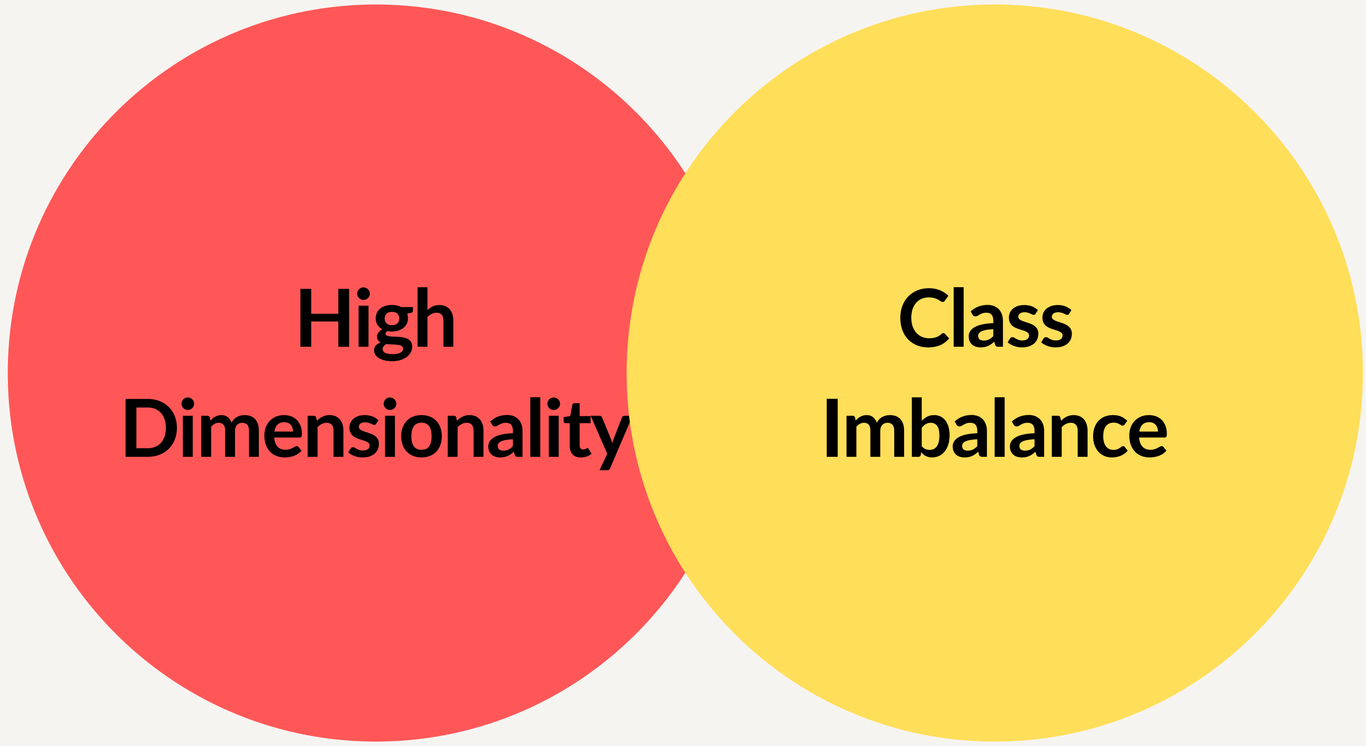
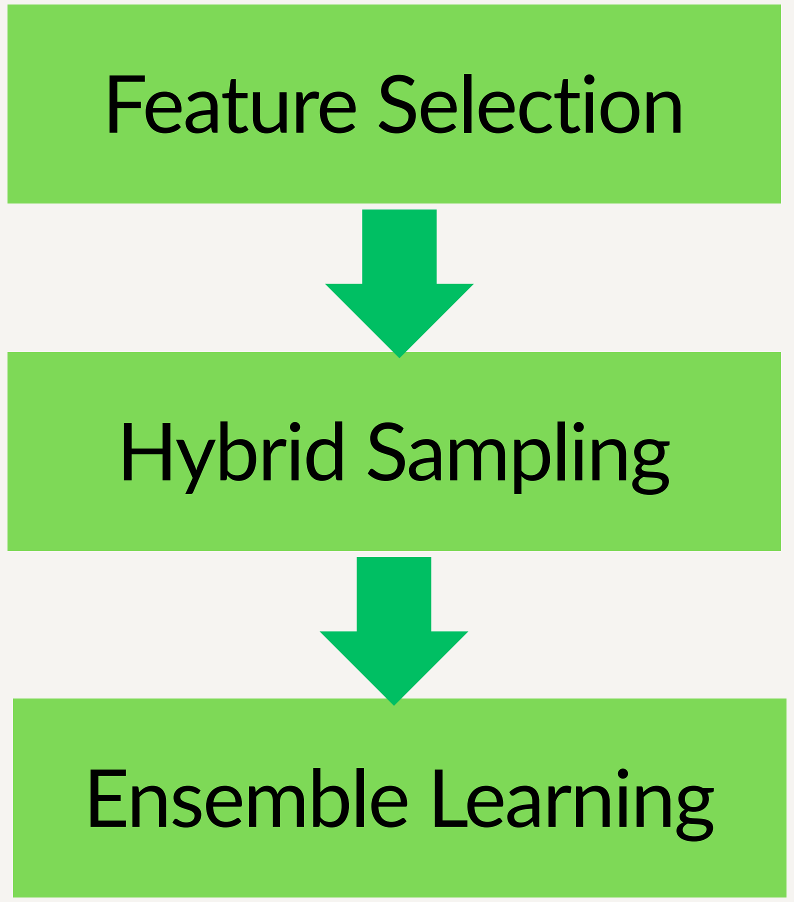
Related Work

Current Research Trends

Technical Issues



Technical Solution



Related Work

A. *Feature Selection and Dimensionality Reduction in NIDS*

Researchers	Feature Selection Method
Nassreddine et al.	Combination of **correlation-based selection** and **XGBoost-based embedded techniques**
Alsaffar et al.	**MI-Boruta (Hybrid)**
Khan et al.	**SelectKBest and correlation reduction**

Related Work

B. Handling Imbalanced Datasets using Hybrid Sampling

Researchers	Feature Selection Method
Hooshmand et al.	SKM-XGB (SMOTE with K-means undersampling)
Khan et al.	SMOTE-Tomek
Nassreddine et al.	SMOTE-Tomek

Related Work

C. Ensemble Learning Approaches for Intrusion Detection

Researchers	Ensemble Learning Method
Farooqi et al.	Voting ensemble (Decision Tree, Random Forest, and XGBoost)
Urmi et al.	Base learner: **Random Forest, XGBoost,** and **Extra-Tree** Meta learner: **Logistic Regression**
Al Essa and Bhaya	Voting and stacking (Random Forest, XGBoost, and MLP)

Related Work

D. Research Gap

- **Lack of Integrated Approaches**: Most studies focus on either feature selection, data sampling, or ensemble learning separately, rather than combining them into a single, comprehensive pipeline.
- **Limited Focus on Multiclass Brute Force Detection**: Few studies specifically investigate how to effectively detect Brute Force attacks within a complex, multiclass classification setting.
- **Challenges with Rare** Attacks: Minority attack classes suffer from limited data samples ("limited attack support"), making them much harder to detect accurately.
- **Unresolved Distribution Shifts**: Cross-domain experiments show that models struggle when the data distribution changes between the training environment and real-world deployment.